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To the Editors, *Quantum Machine Intelligence*

Dear Editors,

I am submitting “SimCert: A Classical-Simulability Audit of Trained Quantum-NLP Text Classifiers” for consideration as a research article in *Quantum Machine Intelligence*.

Quantum natural language processing models are routinely promoted for a quantum benefit on text classification, and that benefit is argued from accuracy. What is rarely tested is whether a trained QNLP circuit ever leaves the efficiently classically-simulable regime. This paper tests it directly.

SimCert re-simulates each trained circuit as a bond-dimension- χ matrix product state, sweeps χ , and reports the smallest value at which a classical surrogate recovers the model’s own predictions. Applied to four published QNLP architectures plus a variational reference, the answer is $\chi = 1$: a product state. On SST-2, the one corpus here whose test set can resolve a few accuracy points, three architectures keep their full accuracy under a product-state surrogate over 200 test items, and a bag-of-words logistic regression matches or beats every audited model on the same data. The required bond dimension stays at 1 as the qubit count grows to sixteen, for two architectures independently, while an exact representation would need 256.

Two controls keep this from being an artifact of the instrument. A positive control with a known required bond dimension recovers that number, so the criterion is not simply printing 1. And auditing along each model’s training path rather than only at its end, the required bond dimension is 1 at 378 of 392 snapshots spanning accuracies from chance to perfect, which rules out the reading that these circuits are cheap only because they failed to learn. One audited model, the compositional DisCoCat pipeline, genuinely needs $\chi = 2$, which is the discriminating case.

The work sits on the journal’s “Quantum Computing for Artificial Intelligence” axis, and it is deliberately constructive rather than dismissive: by naming the quantity a genuine quantum ingredient would have to move, it gives QNLP a target and a reusable instrument. The protocol is model-agnostic, consuming any trained circuit through a common intermediate representation, and the full harness with every stored result is released openly so that any QNLP model can be checked against this bar.

I have tried to be candid about scope. Everything runs at simulator scale on a laptop CPU; the relative-pronoun benchmark turns out not to clear a majority-class predictor and I draw no verdict from it; and the all-quantum mixer runs below its published scale, so I report its audit rather than an accuracy reproduction. None of this settles the asymptotic question, and the paper says so.

This manuscript is original, is not under consideration elsewhere, and has not been published previously. I am the sole author. There are no competing interests and no funding to declare.

I have suggested five reviewers who each published the specific result this work builds on. I have also listed the authors of the four audited architectures as opposed reviewers, since this paper evaluates their published models and reports that a classical surrogate reproduces those models’ predictions.

Thank you for your consideration.

Yours sincerely,

Tanvir Ahmed